



Faculty of Computer Science and Information Technology

Academic Year 2018/2019

Level:1

Course Name: Calculus 1

Course Code: MAT102

Problem Sheet No.:3

Semester: Fall 2018

Instructor: Dr. Rawhy Ismail

Problem 1: Using the limit definition of the derivative, show that $\frac{d}{dx} \cos(x) = -\sin x$ and hence deduce the derivative of $y = \cot(x)$ & $y = \sec(x)$.

Problem 2: Graph the following functions and write the period, domain and range for each one:

- (a) $y = 3 \sin \frac{x}{2}$ (b) $y = 1 - 2 \cos 2x$ (c) $y = |\sin x|$
(d) $y = \tan(x - \frac{\pi}{2})$ (e) $y = \cot x$ (f) $y = \sec x$

Problem 3: Find the derivative of the following functions:

- (a) $y = \sin 3^x$ (b) $y = \sin \sqrt{3 - e^{-x}}$ (c) $y = \sqrt{\cos e^{x^2}}$
✗ (d) $y = \tan(\frac{e^{2x} - e^{-x}}{e^{4x} + 1})$ (e) $y = \sec(\ln(4 - 2x))$ (f) $y = e^{\cot(\frac{1}{x})}$
(g) $y = \ln(e^{\sqrt{\csc(x)}})$ (h) $y = \ln(\csc \sqrt{x})$ (i) $y = e^{\ln(\cos \sqrt{x})}$
(j) $y = \sin(x^\pi \pi^x)$ (k) $y = \frac{1}{\sqrt[3]{\tan x e}}$

Problem 4: Without using a calculator, evaluate:

- (a) $\cos(\tan^{-1} 4) = \dots\dots$
(b) $\cos(2 \sin^{-1} 0.3) = \dots\dots$
(c) $\sin(\tan^{-1} 2 + \cot^{-1} 3) = \dots\dots$

Problem 5: Evaluate in terms of α, β :

(a) $\tan \cos^{-1} \alpha$

(b) $\csc \sin^{-1} \frac{1}{\alpha + \beta}$

✓ (c) $\sec^{-1} \sec \sqrt{\alpha + \beta}$

(d) $\cos(2 \tan^{-1} \beta)$

(e) $\sin(\tan^{-1} \alpha + \cos^{-1} \beta)$

Problem 6: Deduce the derivatives of the following functions and write down the domain:

(a) $y = \cos^{-1} x$

(b) $y = \cot^{-1} x$

(c) $y = \csc^{-1} x$

✓ **Problem 7:** Find the derivative of the following functions:

(a) $y = \sin^{-1} \left(\frac{1 - e^x}{1 + e^{-x}} \right)$

(b) $y = e^{\cos^{-1}(x^4 + 4^x)}$

(c) $y = \sqrt{\tan^{-1} \frac{1}{x}}$

(d) $y = \ln (\cot^{-1} 10^x)$

★ (e) $y = \csc^{-1} \csc \sqrt{x + 1}$

(f) $y = \pi^{\sec^{-1}(x)}$
